Pin Configuration of 8085

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Pin Diagram of 8085 Intel's Microprocessor

The term **microprocessing unit** (MPU) is similar to the term **central processing unit** (CPU) used in traditional computers. We define the MPU as a device or a group of devices (as a unit) that can communicate with peripherals, provide timing signals, direct data flow, and perform computing tasks as specified by the instructions in memory. The unit will have the necessary lines for the address bus, the data bus, and the control signals, and would require only a power supply and a crystal (or equivalent frequency-determining components) to be completely functional.

Using this description, the 8085 microprocessor can almost qualify as an MPU, but with the following two limitations.

1. The low-order address bus of the 8085 microprocessor is **multiplexed** (time-shared) with the data bus. The buses need to be demultiplexed.
2. Appropriate control signals need to be generated to interface memory and I/O with the 8085. (Intel has some specialized memory and I/O devices that do not require such control signals.)

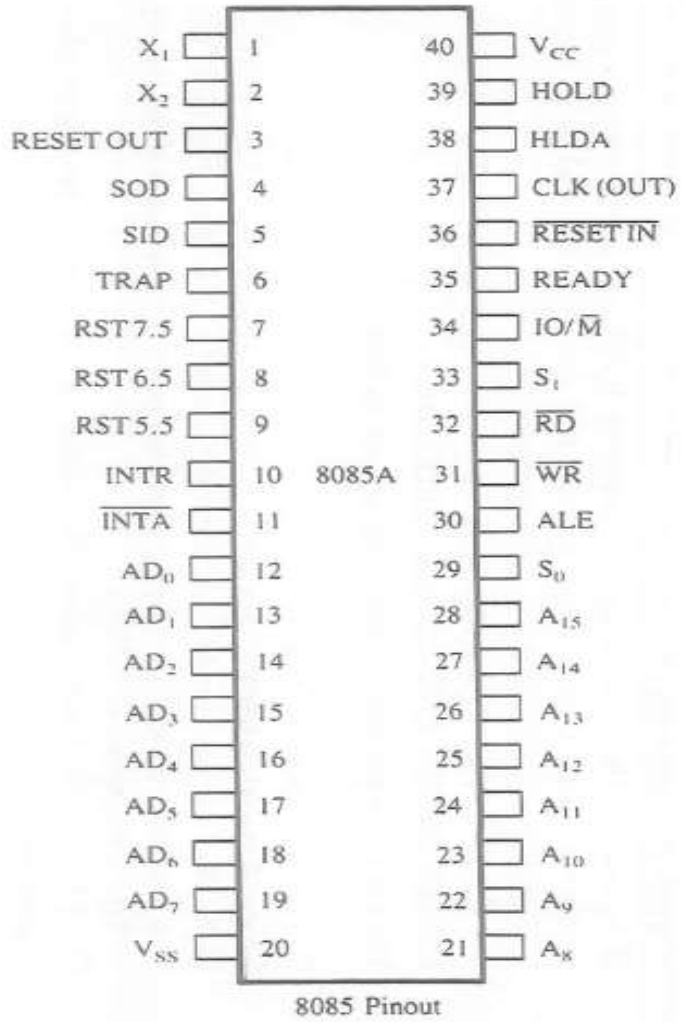
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The 8085 Microprocessor

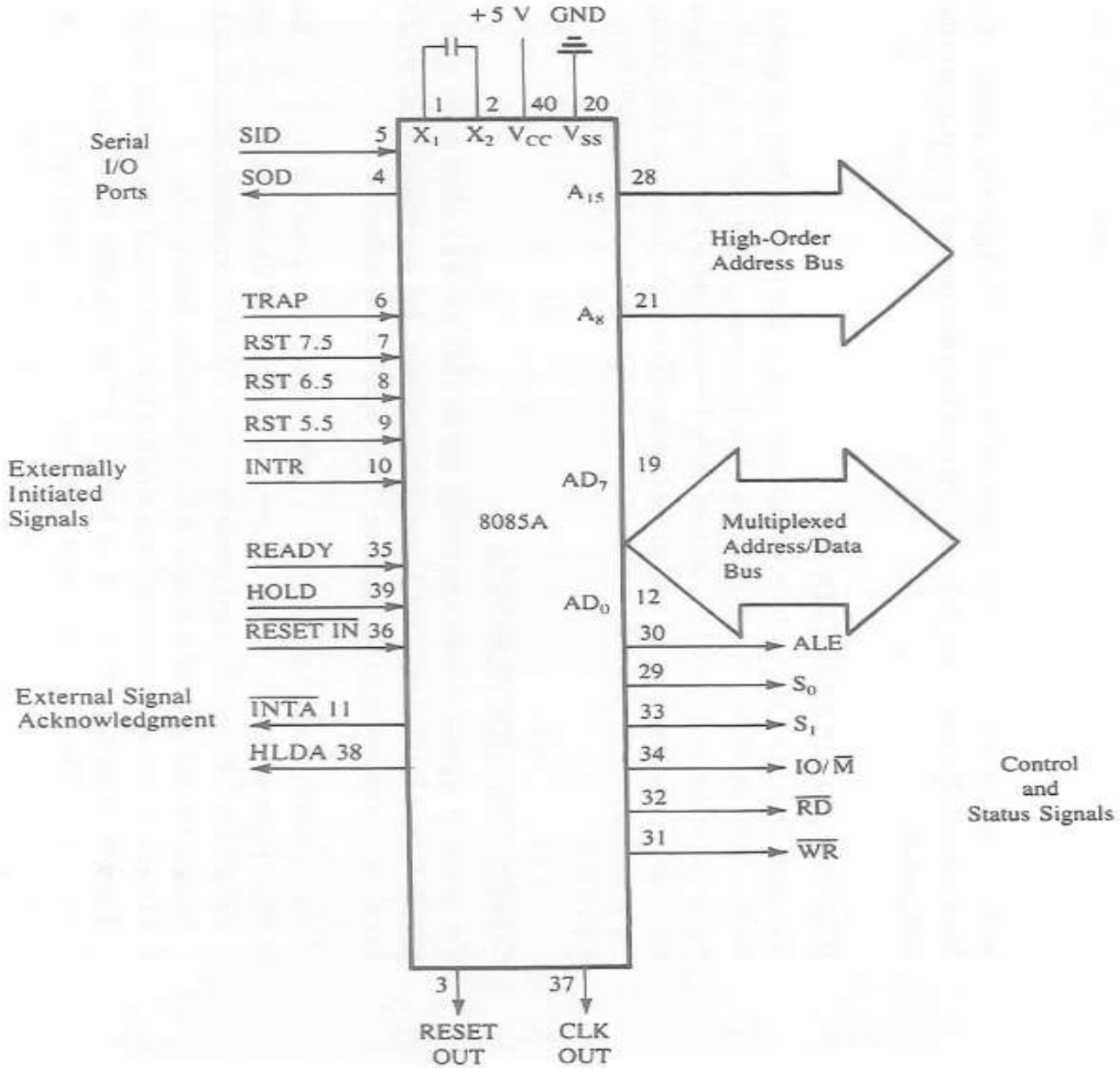
The 8085A (commonly known as the 8085) is an 8-bit general-purpose microprocessor capable of addressing 64K of memory. The device has forty pins, requires a +5 V single power supply, and can operate with a 3-MHz single-phase clock. The 8085A-2 version can operate at the maximum frequency of 5 MHz. The 8085 is an enhanced version of its predecessor, the 8080A; its instruction set is upward-compatible with that of the 8080A, meaning that the 8085 instruction set includes all the 8080A instructions plus some additional ones.

Figure **below** shows the logic pinout of the 8085 microprocessor. All the signals can be classified into six groups: (1) address bus, (2) data bus, (3) control and status signals, (4) power supply and frequency signals, (5) externally initiated signals, and (6) serial I/O ports.

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8085 Pinout



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ADDRESS BUS

The 8085 has 16 signal lines (pins) that are used as the address bus; however, these lines are split into two segments: A_{15}–A_8 and AD_7–AD_0. The eight signal lines, A_{15}–A_8, are unidirectional and used for the most significant bits, called the high-order address, of a 16-bit address. The signal lines AD_7–AD_0 are used for a dual purpose, as explained in the next section.

MULTIPLEXED ADDRESS/DATA BUS

The signal lines AD_7–AD_0 are bidirectional: they serve a dual purpose. They are used as the low-order address bus as well as the data bus. In executing an instruction, during the earlier part of the cycle, these lines are used as the low-order address bus. During the later part of the cycle, these lines are used as the data bus. (This is also known as multiplexing the bus.) However, the low-order address bus can be separated from these signals by using a latch.

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- ALE—Address Latch Enable: This is a positive going pulse generated every time the 8085 begins an operation (machine cycle); it indicates that the bits on AD_7–AD_0 are address bits. This signal is used primarily to latch the low-order address from the multiplexed bus and generate a separate set of eight address lines, A_7–A_0.
- $\overline{RD}$—Read: This is a Read control signal (active low). This signal indicates that the selected I/O or memory device is to be read and data are available on the data bus.
- $\overline{WR}$—Write: This is a Write control signal (active low). This signal indicates that the data on the data bus are to be written into a selected memory or I/O location.
- IO/$\overline{M}$: This is a status signal used to differentiate between I/O and memory operations. When it is high, it indicates an I/O operation; when it is low, it indicates a memory operation. This signal is combined with $\overline{RD}$ (Read) and $\overline{WR}$ (Write) to generate I/O and memory control signals.
- S_1 and S_0: These status signals, similar to IO/$\overline{M}$, can identify various operations, but they are rarely used in small systems. (All the operations and their associated status signals are listed in Table 1 for reference.)

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TABLE 1
8085 Machine Cycle Status and Control Signals

Machine Cycle	Status			Control Signals
	IO/$\overline{M}$	S_1	S_0	
Opcode Fetch	0	1	1	$\overline{RD} = 0$
Memory Read	0	1	0	$\overline{RD} = 0$
Memory Write	0	0	1	$\overline{WR} = 0$
I/O Read	1	1	0	$\overline{RD} = 0$
I/O Write	1	0	1	$\overline{WR} = 0$
Interrupt Acknowledge	1	1	1	$\overline{INTA} = 0$
Halt	Z	0	0	$\overline{RD}$, $\overline{WR}$ = Z and $\overline{INTA} = 1$
Hold	Z	X	X	
Reset	Z	X	X	

NOTE: Z = Tri-state (high impedance)
X = Unspecified

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EXTERNALLY INITIATED SIGNALS, INCLUDING INTERRUPTS

The 8085 has five interrupt signals (see Table 2) that can be used to interrupt a program execution. One of the signals, INTR (Interrupt Request), is identical to the 8080A microprocessor interrupt signal (INT); the others are enhancements to the 8080A. The microprocessor acknowledges an interrupt request by the $\overline{\text{INTA}}$ (Interrupt Acknowledge) signal.

In addition to the interrupts, three pins—RESET, HOLD, and READY—accept the externally initiated signals as inputs. To respond to the HOLD request, the 8085 has one signal called HLDA (Hold Acknowledge).

- $\overline{\text{RESET IN}}$: When the signal on this pin goes low, the program counter is set to zero, the buses are tri-stated, and the MPU is reset.
- RESET OUT: This signal indicates that the MPU is being reset. The signal can be used to reset other devices.

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TABLE 2
8085 Interrupts and Externally Initiated Signals

☐ INTR (Input)	Interrupt Request: This is used as a general-purpose interrupt; it is similar to the INT signal of the 8080A.
☐ $\overline{\text{INTA}}$ (Output)	Interrupt Acknowledge: This is used to acknowledge an interrupt.
☐ RST 7.5 (Inputs) RST 6.5 RST 5.5	Restart Interrupts: These are vectored interrupts that transfer the program control to specific memory locations. They have higher priorities than the INTR interrupt. Among these three, the priority order is 7.5, 6.5, and 5.5.
☐ TRAP (Input)	This is a nonmaskable interrupt and has the highest priority.
☐ HOLD (Input)	This signal indicates that a peripheral such as a DMA (Direct Memory Access) controller is requesting the use of the address and data buses.
☐ HLDA (Output)	Hold Acknowledge: This signal acknowledges the HOLD request.
☐ READY (Input)	This signal is used to delay the microprocessor Read or Write cycles until a slow-responding peripheral is ready to send or accept data. When this signal goes low, the microprocessor waits for an integral number of clock cycles until it goes high.

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SERIAL I/O PORTS

The 8085 has two signals to implement the serial transmission: SID (Serial Input Data) and SOD (Serial Output Data). In serial transmission, data bits are sent over a single line, one bit at a time, such as the transmission over telephone lines.